

Tirth Shah

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EDUCATION

University of Connecticut (GPA: 3.6/4.0)

Master's Degree, Computer Science

Storrs, CT, US

Aug 2024 - May 2026

Coursework: Distributed Systems, Deep Learning, Data Mining, Database Systems, Big Data Algorithms, System Design

Charotar University of Science & Technology (GPA: 3.8/4.0)

Bachelor's Degree, Computer Science

Gujarat, India

Aug 2020 - May 2024

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL, C++, Go, Rust, R, Bash, Shell Scripting

Backend: FastAPI, Node.js, Express.js, Spring Boot, Flask, Django, SQLAlchemy, REST APIs, GraphQL, gRPC

Frontend: React, Next.js, Tailwind CSS, Redux, Vite, HTML5, CSS3, Webpack, Server Components, NextAuth

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, MeiliSearch, Kafka, Airflow, Pinecone, FAISS

Cloud DevOps: Docker, Kubernetes, AWS (EC2, S3, Lambda, RDS), GCP, Terraform, GitHub Actions, CI/CD, Linux

Testing & Observability: pytest, Jest, Playwright, Cypress, Prometheus, Grafana, OpenTelemetry, Structured Logging

Engineering Practices: System Design, OOP, Design Patterns, TDD, JWT, OAuth2, RBAC, Agile Scrum, SDLC, Microservices

AI & ML: scikit-learn, PyTorch, Hugging Face, LangChain, RAG, Vector Search, LLM APIs, OpenAI, Prompt Engineering

EXPERIENCE

Software Engineer – Capital Spirits

Jun 2025 - Mar 2026 | Hartford, CT

- Rebuilt **Next.js** + **TypeScript** catalog search by replacing **Prisma** LIKE queries with **MeiliSearch**, cutting 30,000+ SKU lookups from 4s to 30ms via a tuned ranking pipeline, typo tolerance, faceted filters, and synonym rules.
- Built a **Node.js** batch indexing pipeline syncing **Prisma/PostgreSQL** into **MeiliSearch** with live stock across 3 retail locations, automated facet mapping, and incremental reindex jobs running on a cron schedule.
- Owned the **Stripe Elements** checkout end-to-end with **idempotent webhooks**, server-side price verification, CT 6.35% tax, and duplicate-order guards, eliminating payment disputes across the multi-location storefront.
- Shipped 3-strategy **NextAuth** with Google, Apple, and email login on **JWT sessions**, **OAuth2** provider linking, and a custom **Prisma** adapter, resolving account-linking edge cases on the sign-up flow.

Software Engineer – University of Connecticut

Sep 2024 - May 2026 | Storrs, CT

- Built **Python** automation for accommodation intake, validation, and exam scheduling for 1,000+ students with **Pandas**, **SQL**, and **pytest** unit coverage, cutting manual triage by 40% and replacing legacy spreadsheets.
- Developed **Pandas** + **SQL ETL** workflows with automated data validation and schema checks that blocked malformed records before dashboard refreshes and audit exports, eliminating downstream reconciliation rework.
- Benchmarked **scikit-learn** routing classifiers with precision-recall and confusion-matrix evaluation, lifting routing accuracy 25% and saving \$12k/year in operational cost through reduced misrouting.

Software Engineer – Deconstruct

Jan 2022 - Apr 2024 | Bengaluru, India

- Built an internal **admin and order-management platform** for a D2C skincare brand with **Python**, **Django**, and **React**, centralizing catalog, pricing, inventory, and fulfillment workflows for 500+ SKUs in one panel for internal staff.
- Designed **PostgreSQL**-backed **REST APIs** for order, inventory, and returns flows with **JWT RBAC**, **Pydantic** validation, and audit logging, replacing manual spreadsheet processes across the operations team.
- Shipped **Docker**-packaged services deployed via **GitHub Actions** with **schema validation** and automated tests, cutting back-office data-entry errors and rework 40% across internal tooling.

PROJECTS

AI-Driven Code Generation SAAS Platform (Lovable / v0.dev) | Spring Boot, Spring AI, Kubernetes, Redis, Stripe

- Built a distributed **AI SAAS** platform that generates full **React** apps from natural-language prompts (e.g., “Build a snake game in React”) using **Spring Boot** + **Spring AI**.
- Implemented **SSE** real-time token streaming for 10K+ concurrent sessions at sub-200 ms latency, backed by **MinIO** object storage and **NFS** shared volumes for reliable code persistence.
- Deployed auto-updating build pods on **Kubernetes (Fabric8 + Ingress, < 2s cold start)** with autoscaling for 50K+ codegen requests/day at 99.9% reliability, plus **RBAC**, token quotas, and **Stripe** subscriptions for multi-tenant SAAS.

RoomLock: Concurrent Booking API | FastAPI, PostgreSQL, SQLAlchemy, Alembic, Stripe, Docker | GitHub

- Built a production-style **FastAPI** + **PostgreSQL REST API** with **SQLAlchemy 2.0**, **Alembic** migrations, **Pydantic v2** validation, **JWT RBAC**, and **OpenAPI** docs across reservation, auth, and payment paths.
- Prevented double-booking under concurrent load via **PostgreSQL SELECT FOR UPDATE** row-level locks; modeled **RESERVED**, **CONFIRMED**, **CANCELLED** states with **Stripe webhook** verification and an **APScheduler** reaper for expired holds.

ReflectivePath: Motivational Interviewing Coach | Python, Streamlit, Gemma 2, LoRA, FAISS, LangGraph, Hugging Face | GitHub

- Fine-tuned **Gemma 2-2B** with **LoRA** via **Hugging Face PEFT**, served local **Ollama** inference in **Streamlit**, and routed riskier turns to a **Claude** fallback with prompt-level guardrails and crisis-routing logic.
- Built **FAISS RAG** retrieval over AnnoMI plus synthetic dialogues with **LangGraph** state tracking emotion and defensiveness; evaluated 20 scenarios with **Claude-as-judge** scoring and routed crisis language to 988/SAMHSA resources.

LEADERSHIP

GDSC AI ML Team Lead – Led AI/ML peer learning, project discussions, and developer community support.

Rotaract Club Technical Coordinator – Managed technology operations, event workflows, and digital coordination.